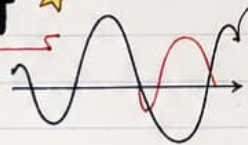




Chapter 7

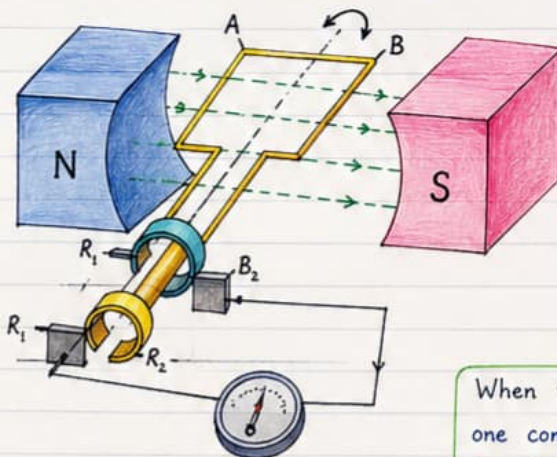
ALTERNATING CURRENT



Alternating Current (AC) is that current whose magnitude and direction change periodically with time.

1. AC GENERATION

When a coil of N turns and area A is rotated with uniform angular speed ω in a uniform magnetic field B , an emf is induced in it. The ends of the coil are connected to two slip rings R_1 and R_2 . The brushes B_1 and B_2 collect the generated emf.



Components

- Coil (ABCD)
- Magnetic field (N to S)
- Slip rings (R_1, R_2)
- Carbon brushes (B_1, B_2)
- External circuit

When the coil rotates one complete revolution, the direction of induced emf reverses once.

2. INSTANTANEOUS EMF

The instantaneous value of emf induced in the rotating coil is

$$e = E_0 \sin \omega t$$

where,

E_0 = Maximum (peak) emf

ω = Angular frequency (rad s^{-1})

t = Time (s)

DERIVATION SNAPSHOT

Magnetic flux $\phi = NBA \cos(\omega t)$

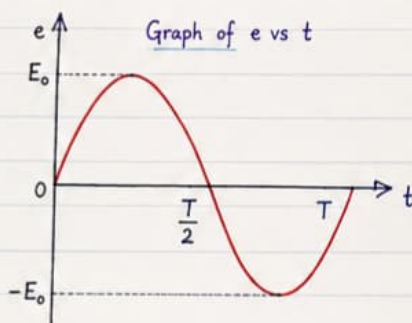
By Faraday's law,

$$e = -N \frac{d\phi}{dt}$$

$$= -N \frac{d}{dt} [NBA \cos(\omega t)]$$

$$= NBA \omega \sin(\omega t)$$

$$E_0 = NBA \omega$$



3. PEAK AND RMS VALUES

Peak (maximum) value

$$E_0 = NBA \omega$$

RMS (effective) value

$$E_{\text{rms}} = \frac{E_0}{\sqrt{2}} = 0.707 E_0$$

Remember

AC produces same heating effect as DC having value E_{rms} .

Similarly for current:

$$i = I_0 \sin \omega t$$

$$I_{\text{rms}} = \frac{I_0}{\sqrt{2}}$$

Average value over one complete cycle = 0

4. TIME PERIOD, FREQUENCY & ANGULAR FREQUENCY

• Time Period (T): Time taken to complete one cycle.

• Frequency (f): Number of cycles per second.

$$f = \frac{1}{T} \text{ (Hz)}$$

• Angular Frequency (ω):

$$\omega = 2\pi f = \frac{2\pi}{T} \text{ (rad s}^{-1}\text{)}$$

• Relation with linear frequency (ν): $\nu = f$

KEY FORMULAS BOX

$$e = E_0 \sin \omega t$$

$$i = I_0 \sin \omega t$$

$$E_0 = NBA \omega$$

$$E_{\text{rms}} = \frac{E_0}{\sqrt{2}}$$

$$I_{\text{rms}} = \frac{I_0}{\sqrt{2}}$$

$$\omega = 2\pi f = \frac{2\pi}{T}$$

$$f = \frac{1}{T}$$

Differentiate

	DC	AC
Magnitude	Constant	Varies with time
Direction	Constant	Changes periodically
Frequency	0	$f (>0)$
Average Value	Possibly non-zero	Zero over one cycle



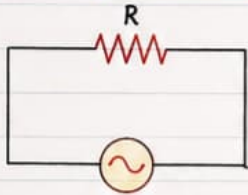
Alternating current is the backbone of modern power systems.





5. AC CIRCUITS : RESISTANCE (R), • INDUCTANCE (L) AND CAPACITANCE (C)

(a) Purely Resistive Circuit (R)



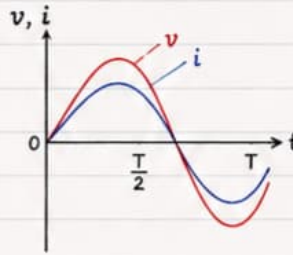
$$v = V_0 \sin \omega t$$

- Only resistance is present.
- Voltage and current are **in phase**.
- Phase difference, $\phi = 0^\circ$

$$v = V_0 \sin \omega t$$

$$i = I_0 \sin \omega t$$

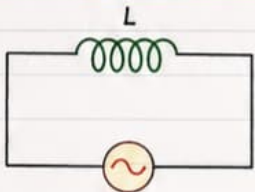
$$V_0 = I_0 R$$



Important Points

- ★ In R circuit, power is always positive.
- ★ Energy is dissipated as heat.
- ★ Average power, $P_{av} = I_{rms}^2 R = V_{rms} I_{rms}$

(b) Purely Inductive Circuit (L)



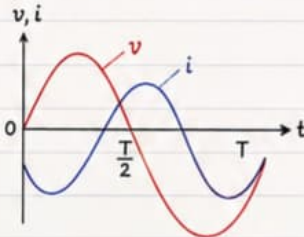
$$v = V_0 \sin \omega t$$

$$v = V_0 \sin \omega t$$

$$i = I_0 \sin \left(\omega t - \frac{\pi}{2} \right)$$

$$V_0 = I_0 X_L$$

- Only inductor is present.
- Current **lags** behind voltage by 90° .
- Phase difference, $\phi = 90^\circ$



Inductive Reactance :

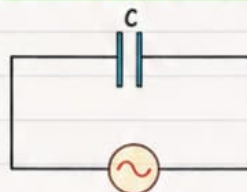
$$X_L = \omega L \text{ (ohm)}$$

ω = Angular frequency (rad s^{-1})

L = Inductance (henry)

Inductor
opposes change
in current.

(c) Purely Capacitive Circuit (C)



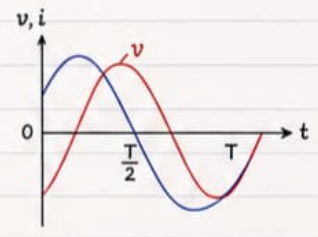
$$v = V_0 \sin \omega t$$

$$v = V_0 \sin \omega t$$

$$i = I_0 \sin \left(\omega t + \frac{\pi}{2} \right)$$

$$V_0 = I_0 X_C$$

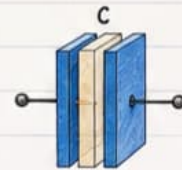
- Only capacitor is present.
- Current **leads** voltage by 90° .
- Phase difference, $\phi = -90^\circ$



Capacitive Reactance :

$$X_C = \frac{1}{\omega C} \text{ (ohm)}$$

C = Capacitance (farad)



Capacitor
opposes change
in voltage.

6. IMPEDANCE (Z) AND AC CIRCUIT WITH R, L, C IN SERIES

Impedance (Z)

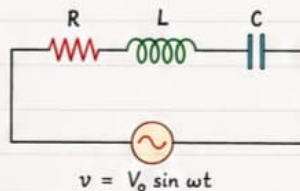
It is the total opposition offered by the circuit to the flow of AC.

$$Z = \frac{V_{rms}}{I_{rms}} \text{ (ohm)}$$

For Pure Circuits :

Circuit	Impedance (Z)	Phase angle(ϕ)	Current
R	$Z = R$	0°	In phase with V
L	$Z = X_L = \omega L$	$+90^\circ$	Lags V by 90°
C	$Z = X_C = \frac{1}{\omega C}$	-90°	Leads V by 90°

(a) Series R-L-C Circuit



$$v = V_0 \sin \omega t$$

Impedance of R-L-C series circuit

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$\tan \phi = \frac{X_L - X_C}{R}$$

ϕ = Phase difference

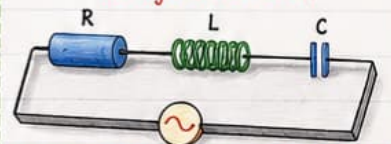
(b) Resonance in R-L-C Circuit

- Resonance occurs when $X_L = X_C$
($\omega L = \frac{1}{\omega C}$)
- Then, $Z = R$ (minimum)
- $\phi = 0^\circ$ and current is maximum.

At Resonance

- ✓ $Z = R$
- ✓ $I_{rms} = \frac{V_{rms}}{R}$ (maximum)
- ✓ Power is maximum
- ✓ Power factor = $\cos \phi = 1$

3D style series circuit



Remember

- Inductor stores energy in magnetic field.
- Capacitor stores energy in electric field.
- In AC, only RMS values are used for power calculations.

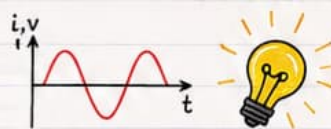
AC is the heart of
modern electrical technology!





Chapter 7

ALTERNATING CURRENT



7. POWER IN AC CIRCUITS

In AC circuits, power is not constant. Hence, average power over one complete cycle is considered.

(a) Instantaneous Power

$$p = vi = (V_0 \sin \omega t)(I_0 \sin(\omega t - \phi))$$

$$= V_0 I_0 [\cos \phi - \cos(2\omega t - \phi)]$$

Power has two parts →

- Constant (dc) part = $V_0 I_0 \cos \phi$
- Variable part = $-V_0 I_0 \cos(2\omega t - \phi)$
(average over a cycle = 0)

(b) Average Power (P_{av})

Average over one complete cycle (T):

$$P_{av} = V_{rms} I_{rms} \cos \phi$$

For pure resistance ($\phi = 0^\circ$)

$$P_{av} = V_{rms} I_{rms} = I_{rms}^2 R = \frac{V_{rms}^2}{R}$$

Meaning of $\cos \phi$

$\cos \phi$ is called the power factor.

It tells how effectively current is converted into useful power.

(c) Power vs Phase angle (ϕ)

Phase angle (ϕ)	Power factor ($\cos \phi$)	Average Power (P_{av})
0°	1	Maximum
$0^\circ < \phi < 90^\circ$	+ve (lagging)	Less than maximum
$\phi = 90^\circ$	0	Zero
$-90^\circ < \phi < 0^\circ$	+ve (leading)	Less than maximum
$\phi = \pm 180^\circ$	-1	Maximum (negative)

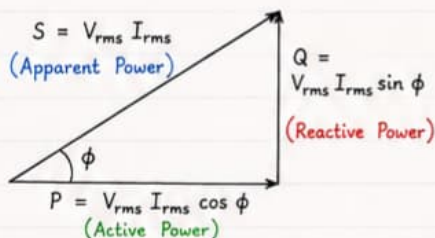
Power Factor ($\cos \phi$)

- For R circuit → $\cos \phi = 1$ (unity)
- For L circuit → $\cos \phi = 0$ (lagging)
- For C circuit → $\cos \phi = 0$ (leading)
- For R-L-C circuit → $0 < \cos \phi < 1$

Remember

Low power factor means more current for same power → more loss in wires and heating.

Power Triangle (for R-L-C series circuit)



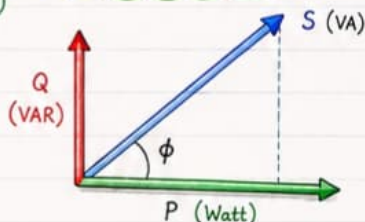
$P \rightarrow$ Active (True) Power (Watt)

$Q \rightarrow$ Reactive Power (VAR)

$S \rightarrow$ Apparent Power (VA)

$$S^2 = P^2 + Q^2$$

$$\tan \phi = \frac{Q}{P}$$



8. RESONANCE IN SERIES R-L-C CIRCUIT (Detailed)

Condition for resonance

$$X_L = X_C$$

or

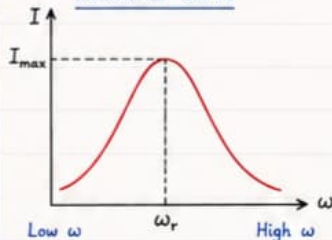
$$\omega L = \frac{1}{\omega C}$$

$$\omega_r = \frac{1}{\sqrt{LC}} \text{ (resonant angular frequency)}$$

At resonance:

- $Z = R$ (minimum)
- Current (I) is maximum
- Power factor $\cos \phi = 1$
- Voltage across L and C are equal in magnitude but opposite in phase ($V_L = V_C$)
- Impedance is purely resistive

Resonance Curve



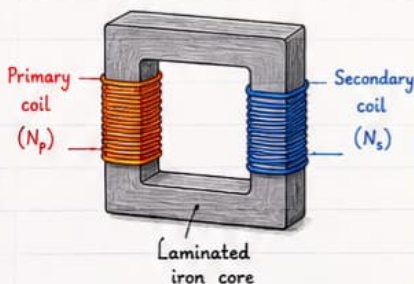
Quality Factor (Q)

$$Q = \frac{\omega_r L}{R} = \frac{1}{\omega_r C R}$$

- Higher Q → sharper & higher resonance peak.
- Bandwidth ($\Delta \omega$) = $\omega_2 - \omega_1$

$$Q = \frac{\omega_r}{\Delta \omega}$$

9. TRANSFORMER (AC Device)



Works on principle of mutual induction. AC in primary produces changing flux, which induces emf in secondary.

Ideal Transformer Relations

$$\frac{V_p}{V_s} = \frac{N_p}{N_s} = \frac{I_s}{I_p} = k$$

$V_p, V_s \rightarrow$ primary & secondary voltages
 $I_p, I_s \rightarrow$ primary & secondary currents
 $N_p, N_s \rightarrow$ turns in primary & secondary
 $k \rightarrow$ turns ratio

Step-Up Transformer

$$N_s > N_p$$

$$V_s > V_p$$

$$I_s < I_p$$



Step-Down Transformer

$$N_s < N_p$$

$$V_s < V_p$$

$$I_s > I_p$$



Applications

- ★ Power transmission (step-up)
- ★ Distribution (step-down)
- ★ Charging adapters
- ★ Electronic devices

AC is everywhere – from power plants to your phone charger!

Understand the concept, not just the formula.





Chapter 7

ALTERNATING CURRENT

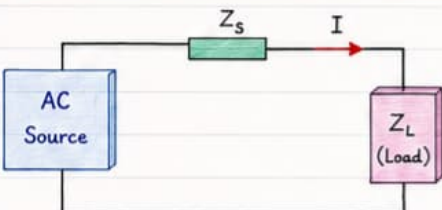


10. MAXIMUM POWER TRANSFER THEOREM (AC)

In an AC circuit, maximum power is delivered to the load when the **impedance of the load** equals the **complex conjugate** of the impedance of the source.

Condition for maximum power:

$$Z_L = Z_S^* \quad \text{where, } Z_S^* = \text{complex conjugate of source impedance}$$



Note

This theorem is very useful in communication systems and audio amplifiers.

12. POWER IN R-L-C CIRCUIT

In a series R-L-C circuit:

★ **Real Power (P)** → Consumed in R (watt)

★ **Reactive Power (Q)** → Exchanged between L and C (VAR)

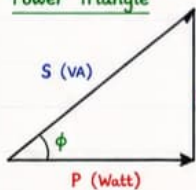
★ **Apparent Power (S)** → Product of V_{rms} and I_{rms} (VA)

$$P = I_{rms}^2 R = V_{rms} I_{rms} \cos \phi$$

$$Q = I_{rms}^2 (X_L - X_C) = V_{rms} I_{rms} \sin \phi$$

$$S = V_{rms} I_{rms} = \sqrt{P^2 + Q^2}$$

Power Triangle



$$\cos \phi = \frac{P}{S} \quad (\text{Power factor})$$

$$\sin \phi = \frac{Q}{S}$$

$$\tan \phi = \frac{Q}{P}$$

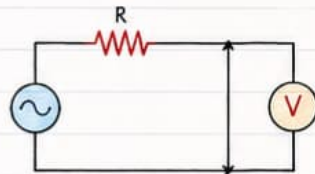
Power factor

It indicates how efficiently electrical power is converted into useful work. Higher $\cos \phi$ → better efficiency.

11. AC VOLTMETER AND AMMETER

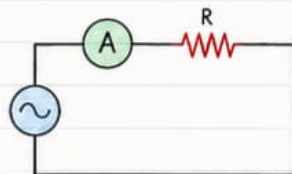
(a) AC Voltmeter

- Connected in parallel across the element.
- Measures RMS voltage.
- Has very high resistance (ideally infinite).



(b) AC Ammeter

- Connected in series in the circuit.
- Measures RMS current.
- Has very low resistance (ideally zero).



RMS (Effective) Reading:

AC voltmeter → reads V_{rms}

AC ammeter → reads I_{rms}

Key idea

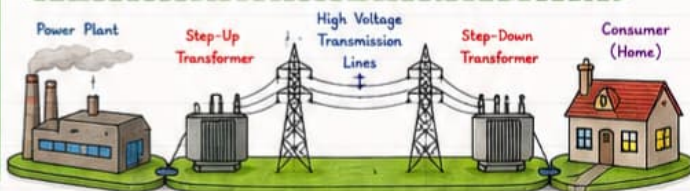
AC instruments are calibrated in RMS values.

13. AC IN DAILY LIFE

- ★ AC is generated in power plants.
- ★ Transmitted over long distances using high voltage (to reduce power loss).
- ★ Then stepped down for domestic use (230 V, 50 Hz in India).
- ★ Used in fans, refrigerators, washing machines, bulbs, motors, heaters, computers, etc.

Advantages of AC over DC

- ✓ Easy and economical to generate.
- ✓ Voltage can be easily stepped up or down using transformers.
- ✓ Efficient transmission over long distances.
- ✓ AC machines are simple, robust and cheaper.



14. IMPORTANT FORMULAS SUMMARY

Quantity	Formula
• AC Generation (emf)	$e = NBA \omega \sin \omega t$
• Peak emf	$E_0 = NBA \omega$
• RMS emf	$E_{rms} = \frac{E_0}{\sqrt{2}} = 0.707 E_0$
• RMS current	$I_{rms} = \frac{I_0}{\sqrt{2}}$
• Time period	$T = \frac{1}{f}$
• Angular frequency	$\omega = 2\pi f = \frac{2\pi}{T}$
• Reactances	$X_L = \omega L, X_C = \frac{1}{\omega C}$
• Impedance (series RLC)	$Z = \sqrt{R^2 + (X_L - X_C)^2}$
• Power (RLC circuit)	$P = V_{rms} I_{rms} \cos \phi$
• Reactive Power	$Q = V_{rms} I_{rms} \sin \phi$
• Apparent Power	$S = V_{rms} I_{rms}$
• Power Factor	$\cos \phi = \frac{P}{S}$

At Resonance

$$(X_L = X_C)$$

- ★ $Z = R$ (minimum)
- ★ $\phi = 0^\circ$
- ★ I_{rms} is maximum
- ★ Power factor = 1



Remember

In AC circuits, always use RMS values for calculations of power.

15. QUICK REVISION

- AC changes magnitude and direction periodically.
- Average value over full cycle = 0.
- RMS value is the effective value.
- In R circuit → V and I in phase.
- In L circuit → I lags V by 90° .
- In C circuit → I leads V by 90° .
- In RLC circuit → Phase angle ϕ depends on $(X_L - X_C)$.
- At resonance → $X_L = X_C, \phi = 0^\circ, Z = R$.
- Power factor ($\cos \phi$) tells efficiency.
- Power = $V_{rms} I_{rms} \cos \phi$.



Understand the concept, not just the formula. Practice more, score more!



Alternating current is not just electricity - it is the **heartbeat** of our modern world.

